

JEN Pricing Proposal

2011 pricing proposal

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Attachment 1 – JEN 2011 network pricing model

Attachment 2 – Schedule of JEN's proposed 2011 tariffs

Attachment 3 – JEN 2011 public lighting model

Attachment 4 – Agreed upon procedures report from KPMG and signed Board of Approval letter

1 Introduction

1.1 Submission purpose

The National Electricity Rules (**the Rules**) rule 6.18.2(a) requires that JEN must submit an initial pricing proposal to the Australian Energy Regulator (**AER**) within 15 business days after publication of the AER's distribution determination.

The AER published its distribution determination on 29 October 2010 and JEN hereby provides its pricing proposal in accordance with rule 6.18.2(a).

1.2 JEN's pricing

JEN has sought to establish efficient tariffs reflecting its different customer bases. In accordance with the Rule requirements, JEN has established its tariff classes and the tariffs it proposes for each of these:

- to ensure that the expected revenue recovered for each tariff class lies on or between stand alone and avoidable cost; and
- having regard to:
 - its estimated long run marginal cost
 - the need to recover its allowed costs in a manner that least distorts efficient consumption patterns
 - end users' ability to respond to price signals
 - transaction costs.

1.3 Submission structure and rule compliance

JEN has structured this submission to demonstrate compliance with each of the requirements of rule 6.18.2(b). The submission dedicates a chapter to each of the key areas of rule compliance:

- Chapter 2 – Tariff classes
- Chapter 3 – Efficient pricing bounds for each DUOS tariff class
- Chapter 4 – Efficient tariffs

- Chapter 5 – Transmission costs, pass throughs and jurisdictional scheme recoveries.

1.3.1 Pricing model

This submission also includes JEN's proposed tariff approval model (**Attachment 1**). In the time available since the AER published its 2011 network tariff approval model on 17 November 2010, JEN has attempted to populate this model. Given the time constraint between 17 and 22 November 2010 submission date, JEN will continue to further review this model and will inform that AER of any issues it identified if and when these arise.

1.3.2 Specific rule compliance

Table 1-1 sets out the specific rule requirement and where JEN's demonstration of compliance can be found in this pricing proposal.

Table 1-1: Rule compliance submission references

Topic	Relevant rules	Submission reference
Pricing proposal elements	6.18.2(b)(1) set out the tariff classes that are to apply for the relevant regulatory year;	Section 2.1
	6.18.2(b)(2) set out the proposed tariffs for each tariff class;	Attachment 1
	6.18.2(b)(3) set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates;	Attachment 2
	6.18.2(b)(4) set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year;	Attachment 1
	6.18.2(b)(5) set out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur;	Section 5.1
	6.18.2(b)(6) set out how charges incurred by the Distribution Network Service Provider for transmission use of system services are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year;	Attachments 1 and 2, and section 5.2

Topic	Relevant rules	Submission reference
	6.18.2(b)(6A) set out how jurisdictional scheme amounts for each approved jurisdictional scheme are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those amounts;	Attachment 1
	6.18.2(b)(6B) describe how each approved jurisdictional scheme that has been amended since the last jurisdictional scheme approval date meets the jurisdictional scheme eligibility criteria;	Section 5.3
	6.18.2(b)(7) demonstrate compliance with the Rules and any applicable distribution determination;	All
	6.18.2(b)(8) describe the nature and extent of change from the previous regulatory year and demonstrate that the changes comply with the Rules and any applicable distribution determination.	Attachment 1
Tariff class principles	6.18.3(a) A pricing proposal must define the tariff classes into which customers for direct control services are divided.	Section 2.1
	6.18.3(b) Each customer for direct control services must be a member of 1 or more tariff classes.	Attachment 1
	6.18.3(c) Separate tariff classes must be constituted for customers to whom standard control services are supplied and customers to whom alternative control services are supplied (but a customer for both standard control services and alternative control services may be a member of 2 or more tariff classes).	Section 2.1
	6.18.3(d) A tariff class must be constituted with regard to: (1) the need to group customers together on an economically efficient basis; and (2) the need to avoid unnecessary transaction costs.	Section 2.2
Pricing principles	6.18.5(a) For each tariff class, the revenue expected to be recovered should lie on or between: (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and (2) a lower bound representing the avoidable cost of not serving those customers.	Chapter 3, Attachments 3 and 4

Topic	Relevant rules	Submission reference
	6.18.5(b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class: (1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and	Chapter 4
	6.18.5(b)(2) must be determined having regard to: (i) transaction costs associated with the tariff or each charging parameter; and (ii) whether customers of the relevant tariff class are able or likely to respond to price signals. (c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.	Chapter 4
Side constraint	6.18.6(a) This clause applies only to tariff classes related to the provision of standard control services. (b) The expected weighted average revenue to be raised from a tariff class for a particular regulatory year of a regulatory control period must not exceed the corresponding expected weighted average revenue for the preceding regulatory year in that regulatory control period by more than the permissible percentage.	Attachment 1
	6.18.6(c) The permissible percentage is the greater of the following: (1) the CPI-X limitation on any increase in the Distribution Network Service Provider's expected weighted average revenue between the two regulatory years plus 2%; Note: The calculation is of the form $(1 + \text{CPI})(1 - X)(1 + 2\%)$ (2) CPI plus 2%. Note: The calculation is of the form $(1 + \text{CPI})(1 + 2\%)$	Attachment 1

Topic	Relevant rules	Submission reference
	6.18.6(d) In deciding whether the permissible percentage has been exceeded in a particular regulatory year, the following are to be disregarded: (1) the recovery of revenue to accommodate a variation to the distribution determination under rule 6.6 or 6.13; (2) the recovery of revenue to accommodate pass through of charges for transmission use of system services to customers; and (3) the recovery of revenue to accommodate pass through of jurisdictional scheme amounts for approved jurisdictional schemes.	Attachment 1
	6.18.6(e) This clause does not, however, limit the extent a tariff for customers with remotely-read interval metering or other similar metering technology may vary according to the time or other circumstances of the customer's usage.	Attachment 1
Transmission tariffs	6.18.7(a) A pricing proposal must provide for tariffs designed to pass on to customers the charges to be incurred by the Distribution Network Service Provider for transmission use of system services.	Attachments 1 and 2
	6.18.7(b) The amount to be passed on to customers for a particular regulatory year must not exceed the estimated amount of the transmission use of system charges for the relevant regulatory year adjusted for over or under recovery in the previous regulatory year.	Attachment 1
	6.18.7(c) The extent of the over or under recovery is the difference between: (1) the amount actually paid by the Distribution Network Service Provider by way of transmission use of system charges in the previous regulatory year; and (2) the amount passed on to customers by way of transmission use of system charges by the Distribution Network Service Provider in the previous regulatory year.	Attachment 1
Jurisdictional scheme	6.18.7A(a) A pricing proposal must provide for tariffs designed to pass on to customers a Distribution Network Service Provider's jurisdictional scheme amounts for approved jurisdictional schemes.	Attachments 1 and 2

Topic	Relevant rules	Submission reference
	(b) The amount to be passed on to customers for a particular regulatory year (year t) must not exceed the estimated amount of jurisdictional scheme amounts for a Distribution Network Service Provider's approved jurisdictional schemes for year t adjusted for over or under recovery in accordance with paragraph 6.18.7(c).	Attachment 1

1.3.3 *Submission values and terminology*

This submission employs the following standards:

- All cost estimates and revenues are expressed in real \$2010 unless otherwise stated
- All prices are expressed in nominal \$2011
- The term 'customer' should be interpreted as an end user of electricity rather than a retailer.

2 **Tariff classes**

In this section JEN sets out its tariff classes for 2011 and demonstrates how it has complied with the relevant rule requirements when establishing these tariff classes.

2.1 **JEN's tariff classes**

2.1.1 *Rule requirements*

Rule 6.18.3 requires that:

- (a) A pricing proposal must define the tariff classes into which customers for direct control services are divided.
- (b) Each customer for direct control services must be a member of 1 or more tariff classes.
- (c) Separate tariff classes must be constituted for customers to whom standard control services are supplied and customers to whom alternative control services are supplied (but a customer for both standard control services and alternative control services may be a member of 2 or more tariff classes).

2.1.2 Distribution use of system services

JEN proposes to retain its existing tariff classes for standard control distribution use of system (**DUOS**) services. Table 2-1 sets out JEN's DUOS tariff classes and the tariffs that are categorised within each of these.

Table 2-1: Tariff classes for standard control DUOS services

Tariff class	Relevant tariffs	Class definition
Residential	A100 / F100 General Purpose A10I / F10I Time of Use Interval Meter A140 Time of Use (closed to new entrants) A180 Off Peak Heating Only	Only available to residential customers
Small business	A200 / F200 General Purpose A210 / F210 Time of Use Weekdays A230 / F230 Time of Use Weekdays – Demand A250 / F250 Time of Use Extended A270 / F270 Time of Use Extended – Demand A290 Unmetered Supply	Only available to customers: a) with annual consumption < 0.4 GWh AND maximum demand < 150 kVA (120 kW); and b) where supply is not taken from an on-site OR dedicated substation
Large business - low voltage	A300 / F300 LV 0.4 - 0.8 GWh A30E LV _{EN} Annual Consumption 0.8 GWh A320 LV 0.8+ - 2.2 GWh A32E LV _{EN} 0.8+ - 2.2 GWh A340 LV 2.2+ - 6.0 GWh A34E LV _{EN} 2.2+ - 6.0 GWh A34M LV _{MS} 2.2+ - 6.0 GWh A370 LV 6.0+ GWh A37E LV _{EN} 6.0+ GWh A37M LV _{MS} 6.0+ GWh	Low voltage tariffs (nominal voltage < 1000 volts) Only available to customers: a) with annual consumption >= 0.4 GWh <u>or</u> maximum demand >= 150 kVA (120 kW); or b) taking supply from an on-site <u>or</u> dedicated substation

Tariff class	Relevant tariffs	Class definition
Large business - high voltage	A400 HV A40E HV _{EN} Annual Consumption 10 GWh A40R HV _{RF} (closed to new entrants) A42E HV _{EN} 10+ GWh A480 HV - Annual Consumption >= 55 GWh	High Voltage Tariffs (nominal voltage >= 1000 volts AND <= 22,000 volts)
Large business - sub-transmission	A500 Sub-transmission A50A Sub-transmission MA A50E Sub-transmission EG	Sub-transmission Tariffs (nominal voltage > 22,000 volts)

2.1.3 Alternative control services

In addition to DUOS services, JEN provides a range of alternative control distribution services, which were previously classified as 'excluded services' by the Essential Services Commission of Victoria.

The AER has classified these services into fee based services, quoted services and public lighting services. JEN therefore provides these within three separate tariff classes that correspond to the form of alternative control and service determined by the AER:

- *Fee based services* – Services for which costs are generally discernable prior to undertaking the service and do not vary significantly among customers. For fee based services, the AER has applied a cap on the price per service. Refer to Attachment 3 for details.
- *Quoted services* – Variable services that depend on the particulars of the service JEN provides. For quoted services, the AER has placed a cap on the applicable labour rates (inclusive of margins and all overheads). The labour rates can be applied to quoted service works as appropriate. Materials for quoted services are to be recovered at cost. Refer to Attachment 3 for details.
- *Public lighting services* – Public lighting services relate to the operation, maintenance and repair and replacement (OMR) of public lighting. The AER has applied a cap on the prices of individual public lights. Refer to Attachment 4 for details.

Table 2-2 sets out the fee based and quoted service groupings of alternative control services in accordance with the JEN's distribution determination 2011-15, October 2010.

Table 2-2: Tariff classes for alternative control services

Tariff class	Relevant services	Class definition
Fee based services	Manual energisation of new premises (fuse insert) Manual re-energisation of existing premises (fuse insert) Manual de-energisation of existing premises (fuse removal) Temporary disconnect – reconnect for non-payment Adjust time switch Manual special meter read Connection – temporary supply (overhead supply with coincident abolishment) Service vehicle visits Wasted service vehicle visit (not DNSP fault) Fault response (not DNSP fault) Retest of types 5 and 6 metering installations for first tier customers < 160 MWh Retest of types 5 and 6 metering installations for first tier customers > 160 MWh Reserve feeder Public lighting <i>Routine new connections where JEN is the responsible for metering customers < 100 amps</i> Connection – single phase service connection to new premises Connection – three phase service connection to new premises with direct connected metering <i>Routine new connections where JEN is not the responsible for metering customers < 100 amps</i> Connection – single phase service connection to new premises Connection – three phase service connection to new premises with direct connected metering	Services for which the AER has applied a cap on the price per service.

Tariff class	Relevant services	Class definition
Quoted services	Routine new connections for customers requiring greater than 100 amps including current transformers (CTs) Temporary covering of low voltage mains and service lines Elective undergrounding where an existing overhead service exists High load escorts—lifting of overhead lines Restoration of overhead service cables pulled down by transport vehicles transporting high loads Supply abolishment Rearrangement of network assets at customer request, excluding alteration and relocation of existing public lighting services	Services for which the AER has placed a cap on the applicable labour rates (inclusive of margins and all overheads).

Tariff class	Relevant services	Class definition
Public lighting	Mercury vapour 80 watt Sodium high pressure 150 watt Sodium high pressure 250 watt T5 2x14 watt Fluorescent 20 watt Fluorescent 40 watt Fluorescent 80 watt Mercury vapour 50 watt Mercury vapour 125 watt Mercury vapour 250 watt Mercury vapour 400 watt Sodium low pressure 90 watt Sodium high pressure 50 watt Sodium high pressure 100 watt Sodium high pressure 400 watt Sodium high pressure 250 watt (24 hours) Metal halide 70 watt Metal halide 100 watt Metal halide 150 watt Metal halide 250 watt Incandescent 55 watt Incandescent 100 watt Incandescent 150 watt T5 2X24 Watt 32W Compact Fluorescent 42W Compact Fluorescent	Services for public lighting for which the AER has applied a cap on the price per lighting type.

2.2 Setting efficient tariff classes

2.2.1 Rule compliance

Rule 6.18.3(d) requires that:

A tariff class must be constituted with regard to:

- (1) the need to group customers together on an economically efficient basis; and
- (2) the need to avoid unnecessary transaction costs.

2.2.2 *Economically efficient customer groupings*

JEN's existing five DUOS tariff classes enable it to achieve an optimal balance of differentiated price signalling, taking into account customer characteristics and the transaction costs of providing customised tariffs at a more disaggregated level.

JEN does not see benefit in further tariff class disaggregation as the five tariff classes correspond to JEN's five material customer segments for whom it is necessary to charge differentiated pricing structures and charging parameters.

To the extent that further less material pricing differentiation is desirable within JEN's market segments can be achieved within the pricing flexibility that the NER provides within tariff classes. For example, within the residential tariff class, a DNSP can apply time of use (**TOU**) and flat rate pricing to customers with or without an interval meter respectively. This is possible through the rule 6.18.6 provision that allows tariff rebalancing within the total revenue constraint on a given tariff class.

2.2.3 *Avoiding unnecessary transaction costs*

In developing its tariff classes and its tariffs, JEN has had regard to the need to avoid unnecessary transactions costs that additional tariff classes may impose on JEN, retailers and customers.

This is the primary reason for JEN establishing a single tariff class for residential customers as the transaction costs on retailers and households do not warrant further disaggregation.

JEN has also quantified the transaction costs it faces when introducing new tariffs. JEN estimates these to be in the order of **{c-i-c}** as discussed further in section 4.3.1.

3 **Efficient pricing bounds**

3.1 **Rule requirements**

Rule 6.18.5 requires that revenues from each tariff class for standard control distribution services must lie between economically efficient bounds, specifically:

- (a) For each tariff class, the revenue expected to be recovered should lie on or between:
 - (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and

- 
- (2) a lower bound representing the avoidable cost of not serving those customers.

The purpose of applying stand alone and avoidable cost bounds on expected tariff class revenues is to ensure that, for each tariff class, the DNSP is not pricing outside the bounds defined by economic efficiency. These stand alone and avoidable cost bounds are the highest and lowest theoretical prices that a distributor could charge a customer class without imposing costs on other classes. That is, pricing outside these efficient bounds implies cross subsidisation between customer classes if the business is recovering its costs.

3.2 Stand alone costs

3.2.1 JEN stand alone cost estimation

Stand alone cost represents the cost that would be required to replicate or bypass the network. It follows that if customers were charged above stand alone costs, it would be beneficial for that group of customers to bypass the network, or to be provided by a new entrant, if entry is feasible. Therefore, these costs are comprised of the assets and operating costs that would be required to provide services to that tariff class.

JEN has estimated its stand alone costs for each tariff class using a comprehensive standalone cost estimation model. This model estimates the total capital and operating costs required to provide distribution services to a particular tariff class exclusively. The capital costs are estimated using optimised replacement value for the assets required by each tariff class.

Once these costs are estimated, JEN then calculates the return on assets, return of assets and operating costs in relation to providing such services.

Steps JEN has modelled are as described below.

Step one

The first step involves JEN determining the portion of notional standalone assets required to service a particular tariff class exclusively. The proportion of assets (by asset class) is allocated to the various tariff classes. For example, the proportion of overhead lines pertinent to the various classes of customers is set out in the following table. JEN repeated this approach of allocating standalone asset equivalents to each tariff class for all the asset categories set out in Table 3-1.

Table 3-1: Asset shares by tariff class

Asset type	Residential	Small B	Large B - LV	Large B - HV	Large B – Sub-trans
<i>Sub-transmission</i>					
overhead lines	60%	50%	60%	50%	23%
UG lines	60%	50%	60%	50%	23%
stations	40%	21%	25%	31%	0%
communications	60%	20%	25%	22%	3%
<i>Distribution</i>					
HV poles	31%	21%	25%	23%	0%
LV poles	60%	40%	0%	0%	0%
LV PTS	60%	40%	0%	0%	0%
HV PTS (wood)	31%	21%	25%	23%	0%
HV PTS (conc)	31%	21%	25%	23%	0%
HV PTS (SWER)	100%	0%	0%	0%	0%
HV OH cond	31%	21%	25%	23%	0%
LV OH cond	60%	40%	0%	0%	0%
HV UG cable	31%	40%	25%	23%	0%
LV UG cable	60%	40%	0%	0%	0%
HV metering	0%	0%	0%	100%	0%
substations	60%	30%	25%	0%	0%
HV switchgear	80%	20%	0%	0%	0%
service cable	40%	60%	0%	0%	0%
other dist equip	80%	9%	0%	0%	0%
SCADA/telemetry	100%	100%	100%	100%	100%
<i>Non-network Assets</i>					
distribution	60%	100%	100%	100%	10%
misc. distribution land	60%	100%	100%	100%	10%
sub-transmission	60%	100%	100%	100%	10%

Step two:

The second step involves JEN calculating the value of assets required to provide distribution service to each tariff classes exclusively. Using the percentages

derived in step one, JEN applies these to the optimised replacement value of distribution assets.

Step three:

The third step involves JEN calculating the capital costs required to service the tariff class exclusively. This involves calculating the return on and return of assets attributable to the portion of distribution assets:

- *Return on asset* – a pre-tax nominal vanilla rate of return of 9.95 per cent sourced from the AER's final determination for JEN is applied to the applicable portion of distribution assets.
- *Return of asset* – calculated based on the useful lives of the relevant asset classes.

Step four:

The fourth step requires JEN to determine the proportion of operating and maintenance expenditure (opex) applicable to provide standard distribution services to the tariff classes exclusively. JEN performed this estimation using engineering assessment of opex required for each tariff class on a standalone basis having regard to a scaled rate of fixed costs and applicable customer numbers in each tariff class for variable opex costs.

Step five:

Finally, JEN derives the per kWh costs (capital and operating) for each tariff class. This involves JEN summing the notional building block cost elements for each tariff class, then dividing this by the energy (in kWh) that JEN anticipates it will distribute to that tariff class.

3.2.2 Stand alone cost estimates

Table 3-2 presents the results for each tariff class. It can be observed that the estimate of stand alone costs exceeds the expected revenue for each tariff class.

Table 3-2: Stand alone costs compared to expected revenue (cents/kWh)

Tariff class	Stand alone estimate	Expected revenue
Residential	11.09	6.21
Small business	9.49	5.95
Large business - low voltage	7.29	3.48
Large business - high voltage	9.82	2.00

Tariff class	Stand alone estimate	Expected revenue
Large business - sub-transmission	1.46	0.38

3.3 Avoidable costs

Avoidable cost represents the cost that would be avoided if the DNSP no longer provided services to that group of customers (or 'tariff class'). If the business charges less than avoidable cost to that group of customers, it follows that it would be beneficial for it not to provide services to those customers, since the costs would be greater than the expected revenues.

3.3.1 JEN avoidable cost estimation

JEN has two broad categories of costs that are avoidable through ceasing DUOS services to a given tariff class:

1. *Avoidable capital costs* – The future load related capital expenditure that can be avoided if the service were not provided. Specifically, some network reinforcement costs may be deferred if peak demand were reduced.
2. *Avoidable operating costs* – The customer specific and load related operating costs that can be avoided if the service were not provided.

Avoidable capital costs

To calculate the capital related avoidable costs JEN:

- Estimated each tariff class's contribution to system peak demand
- Estimated the future peak demand for period 2011 to 2050, using a 2 per cent per year growth rate for years beyond the 2011-15 regulatory control period and NIEIR forecasts for the 2011-15 regulatory control period
- Calculated the implied capital requirement for each tariff class, where the components required for this calculation included:
 - a. Average capex per MW of peak growth
 - b. Base Case Load related capex
 - c. Revised capex requirement if tariff removed for each tariff class
- Divided the present value of the implied capital requirement for each tariff class by the energy consumption of that tariff class.

Avoidable operating costs

For avoidable operating costs, JEN identified that a portion of the following operating costs would be avoidable if a given tariff class were no longer supplied:

- Network operating costs
- Billing and revenue collection costs
- Advertising
- Customer service
- Other operating costs.

Network maintenance and regulatory compliance costs are not considered to be avoidable for any given tariff class.

JEN analysed each of the identified avoidable operating cost categories based on the following cost drivers:

- Fixed cost
- Customer numbers
- Unit sales

JEN estimated the proportion of cost relevant to each of these cost drivers. JEN then divided the results by the average forecast energy consumption for the relevant tariff class.

3.3.2 Avoidable cost estimates

Table 3-3 presents the results for each tariff class. It can be observed that the expected revenue for each tariff class exceeds the estimate of avoidable costs.

Table 3-3: Avoidable costs compared to expected revenue (¢/kWh)

Tariff class	Avoidable estimate	Expected revenue
Residential	1.425	6.208
Small business	1.080	5.947
Large business - low voltage	{c-i-c}	3.482
Large business - high voltage	{c-i-c}	2.001

Tariff class	Avoidable estimate	Expected revenue
Large business – sub-transmission	{c-i-c}	0.380

3.4 Summary of Rule compliance

The efficient bounds for each tariff class are presented in Table 3-4.

Table 3-4: Avoidable costs compared to expected revenue (¢/kWh)

Tariff class	Avoidable estimate	Expected revenue (average price)	Stand alone estimate	Point between efficient pricing bounds
Residential	1.425	6.208	11.093	49%
Small business	1.080	5.947	9.487	58%
Large business - low voltage	{c-i-c}	3.482	7.290	{c-i-c}
Large business - high voltage	{c-i-c}	2.001	9.817	{c-i-c}
Large business – sub-transmission	{c-i-c}	0.380	1.457	{c-i-c}

4 Pricing parameters and tariffs

4.1 Rule requirements

The Rules include certain pricing efficiency and cost recovery principles that JEN has had regard to when setting its DUOS tariffs. Specifically, Rule 6.18.5(b) requires that:

- (b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class:
 - (1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and
 - (2) must be determined having regard to:
 - (i) transaction costs associated with the tariff or each charging parameter; and
 - (ii) whether customers of the relevant tariff class are able or likely to respond to price signals.

The Rules also recognise that the building block costs allowed in an AER determination will provide revenues that are greater than LRMC. This is because they include recovery of a business's sunk costs in the form of the return on and of the DNSP's regulatory asset base (**RAB**). This means DNSPs must actually price to recover their long run average cost (**LRAC**).

On account of this, the Rules require DNSPs to price in a manner that least distorts customer's usage decisions. Specifically Rule 6.18.5(c) requires:

- (c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

Application of LRMC

The purpose of requiring tariffs and tariff parameters to be set by taking into account long run marginal costs reflects the economic principle that prices should reflect the underlying costs of providing the service. As consumption increases the capacity of the network requires augmentation to accommodate the additional demand. Therefore in order for customers' consumption decisions to take into



account these increased costs current prices need to reflect the expected additional costs arising from additional consumption.

While JEN has had regard to pricing in line with its LRMC, the Rules also identify specific factors that create a tension affecting the applicability of LRMC to JEN's DUOS service pricing:

- transaction costs can affect the relative costs and benefits of LRMC pricing
- customers may not be able to respond to price signals.

These factors affect the application of LRMC to signal the impact of incremental consumption since often in an electricity network the pricing objective is to flatten the consumption profile rather than increase or decrease consumption per se.

Furthermore, since the building blocks revenue is greater than LRMC (it is long run average cost including return on sunk costs) not every tariff class and tariff parameter can be set with reference to LRMC and it would not be appropriate to do so. This principle is envisaged by NER rule 6.18.5(c).

Since LRMC attempts to capture the change in costs in response to a change in demand, JEN has primarily had regard to its consumption and capacity based charging parameters in JEN's LRMC analysis. Other components of the JEN's tariffs such as its standard charges are not priced with respect to demand and can essentially be viewed as recovering historic costs (such as the return on sunk assets). For this reason, and fact that the building block cost allowance reflects LRAC, JEN's average revenues can be expected to exceed LRMC in most cases.

4.2 Long run marginal cost

Marginal costs represent the change in costs that arise from a change in demand. The types of costs that are captured are differentiated based on the time horizon that is under consideration, that is, whether it is the 'short run' or 'long run'. In the short run, investments in capacity and overhead is fixed and so marginal cost captures operational inputs such as additional labour, materials and energy. However over the long run all inputs can feasibly be altered such that marginal cost captures the cost of building additional capacity.

Marginal costs are essentially forward looking, since they reflect the expected change in costs that arise from changes in demand. Because they are forward looking invariably the estimates are subjective and are best viewed as a range.

4.2.1 *LRMC estimation*

There are two commonly known approaches for estimating LRMC: the Turvey approach; and the average incremental cost (AIC) approach. The Turvey approach aims to capture the direct change in expenditure resulting from a change in demand whereas the AIC approach captures the average change in expenditure. For this reason the AIC approach is more readily applied and so for the purposes of this analysis JEN has utilised the AIC approach.

The AIC approach dictates that an optimal least cost capital programme and associated operating costs be forecast to meet additional demand over a medium term (20 to 30 years). JEN has identified the capex associated with the expanding its network capacity to accommodate changes in demand. JEN has estimated annual incremental operating cost. These combined costs are then divided by the change in demand as forecasted by NIEIR to obtain a per unit estimate of LRMC.

JEN calculated the long run marginal cost estimates using the AIC approach. The cost comprised incremental capex and long run opex. These were divided by incremental demand/volumes.

JEN scrutinised its capex programme and identified those projects which are of an expansionary nature. JEN did this by identifying those projects that were demand related or were reinforcements to the network. JEN then identified those projects which were entirely expansionary in nature and those that augmented existing assets.

JEN allocated the capex to the tariff classes based on peak demand and customer numbers per tariff class as these measures that most closely reflect the requirement for capex.

As part of its calculation of avoidable costs JEN identified those costs associated with growth in its network. JEN calculated the long run opex by multiplying the annual opex identified by the weighted average life of the assets to align the two aspects of cost.

JEN calculated incremental demand in two ways, for large business customers and sub-transmission customers it was calculated by the difference in the peak demand forecast (kW). For residential and small business customers it reflects the annual change in volumes (kWh).

4.2.2 *JEN's LRMC estimates*

Table 4-1 sets out the LRMC estimates JEN has developed using the methodology set out above and which JEN has had regard to when setting its tariffs, in conjunction with the other relevant Rule factors discussed in section 4.3.

Table 4-1: JEN long run margin cost estimates

Tariff class	Unit	LRMC
Residential	¢/kWh	6.95
Small business	¢/kWh	3.95
Large business - low voltage	\$/kW	93.25
Large business - high voltage	\$/kW	92.17
Large business – sub-transmission	\$/kW	92.08

4.3 Other relevant pricing principles

As dictated by the Rules and for the reasons discussed in section 4.1, JEN has had regard to a number of other relevant pricing principles.

4.3.1 Transaction costs

In developing its tariff tariffs, JEN has had regard to the need to avoid unnecessary transactions costs that additional tariffs and charging parameters may impose on JEN, retailers and customers. In doing so, JEN has quantified the transaction costs it faces when introducing new tariffs.

JEN estimates that introducing a new four-part time of use tariff, comprising a standing charge, peak, shoulder and off peak energy components would cost in the order of {c-i-c}. This comprises:

- carrying out the necessary changes in AMI SAP accounting system - {c-i-c}
- carrying out the necessary changes in CIS Plus customer billing system - {c-i-c}
- developing new end of month reports to allow reporting financial performance to the business as well as to demonstrate compliance with JEN regulatory obligations for data maintenance and provision - {c-i-c}.

Further, JEN understands that retailers incur costs to implement new TOU tariffs within their billing systems and to notify customer accordingly.

4.3.2 Customers' ability to respond to price signals

JEN has developed its proposed DUOS tariffs and charging parameters having regard to the following factors that affect customers' ability to respond to price signals:

- retailers may not pass network pricing signals through to customers—for example, retailers may package network prices such that final energy prices peak at different times to network prices, such that network price signals are diluted
- customer may not receive the price signal in a timely manner or understand it in manner that customers to effectively modify their behaviour—for example, because billing is quarterly or because charges are not disaggregated into network and non-network components
- specific customer groups may be unable to respond to price signals, including low income earners and business customers with budgetary constraints (for example, with respect to obtaining systems capable of responding to TOU pricing).¹

4.3.3 *Recovery of approved building block revenues*

The AER's final determination determined JEN's allowed building block revenues for each year of the 2011-15 regulatory control period. It also determined the NPV smoothed price path for recovery of these. This price path required an initial increase in JEN's revenues in 2011 relative to 2010 of 4.99 per cent.² Table 4-2 demonstrates how JEN's proposed 2011 DUOS tariffs provide a compliant revenue increase of 4.99 per cent. Attachment 1 details the calculations that prove this compliance.

¹ Many of these issues were also raised in stakeholder submissions to the AER's price review. See AER, *Draft Decision*, June 2010, p. 145.

² AER, *Final Jemena Electricity Networks (Victoria) Ltd Distribution determination 2011-2015*, 29 October 2010, Table 6.

Table 4-2: Recovery of building block revenues

Item	Unit	2010	2011	2012	2013	2014	2015	NPV
<i>Building block determination</i>								
Building block requirement	\$M		188.2	189.3	205.3	207.2	204.3	751.2
Smoothed forecast revenue	\$M	168.8	179.8	190.1	199.3	209.1	220.8	751.2
Price path or X factor	%		-4.99	-3.00	-3.00	-3.00	-3.00	
<i>Price control</i>								
Notional 2010 revenue (Pt-1 x Qt-2)	\$M	170.4						
Notional 2011 revenue (Pt x Qt-2)	\$M		183.9					
Allowed price movement = $(1+CPI_t)(1-X_t)(1+L_t)$	%	7.95						
Compliant nominal price movement	%		7.95					

5 Transmission costs, pass throughs and jurisdictional scheme recoveries

5.1 Tariff variation for pass throughs

5.1.1 Rule requirements

Rule 6.18.2(b)(5) requires that a DNSP's pricing proposal must:

set out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur

5.1.2 Potential tariff variation for pass throughs

Possible pass through events

Chapter 10 of the Rules specifies that the following pass through events are applicable to all distribution determinations:

- regulatory change event
- a service standard event
- a tax change event
- a terrorism event.

In addition to the pass through events and provisions set out in the Rule, the AER has determined the following pass through events are also applicable to JEN:

- a insurance event
- an insurer credit risk event
- a natural disaster event
- a declared retailer of last resort event
- a network charge pass through event.³

To the extent any of these events transpire during 2011, JEN will apply to the AER for pass through.

³ AER, *Final Jemena Electricity Networks (Victoria) Ltd Distribution determination 2011-2015*, October 2010, section 4.



At present, the only event that JEN is certain will transpire is the network charge pass through event. JEN will provide its pass through application for this event to the AER in accordance with the requirements of the final determination and will adjust its proposed 2012 tariff to recover any amount approved in that application.

Manner of pass through

JEN will assess any pass through event that occurs and recover this from its tariff classes and tariffs in a cost reflective manner having regard to extent to which different customers contributed to JEN incurring the event costs and the other pricing principles in rules 6.18.5(b) and (c).

5.2 Transmission use of system recovery

5.2.1 Rule requirements

Rule 6.18.2(b)(6) requires that a DNSP's pricing proposal must:

set out how charges incurred by the Distribution Network Service Provider for transmission use of system services are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year

5.2.2 Transmission use of system tariffs

JEN has set out a schedule of its proposed TUOS tariffs in Attachment 2. These tariffs are set to recover JEN's required transmission revenues as calculated in accordance with the maximum transmission revenue formula (MTR_t).

MTR_t is specified in section F.2.2 of Appendix F of the Victorian Electricity Distribution Network Service Providers Distribution Determination 2011-2015. This formula includes the calculation requirements for under and over recovery amounts from the two preceding years.

Attachment 1 provides the calculations demonstrating JEN's MTR_t compliance.

5.3 Jurisdictional scheme recoveries

5.3.1 Rule requirements

Rule 6.18.2(b)(6A) requires that a DNSP's pricing proposal must:

(6A) set out how jurisdictional scheme amounts for each approved jurisdictional scheme are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those amounts; and

(6B) describe how each approved jurisdictional scheme that has been amended since the last jurisdictional scheme approval date meets the jurisdictional scheme eligibility criteria

5.3.2 Relevant jurisdictional scheme

The Australian Energy Markets Commission rule change determination in relation to feed-in-tariff schemes that took effect from 1 July 2010. This provides for JEN to recover the payments it is obliged to make under the Electricity Industry Amendment (Premium Solar Feed in Tariff) Act 2009 (Vic). This act requires JEN to provide a premium feed-in tariff (**PFIT**) for photovoltaic installations.

5.3.3 Jurisdictional scheme tariffs

JEN has set out a schedule of its proposed tariffs to recover costs incurred through relevant jurisdiction schemes in Attachment 2. These tariffs are set to recover JEN's required jurisdictional scheme revenues as calculated in accordance with the jurisdictional scheme revenue formula (**MJR_t**).

MJR_t is specified in section F.2.3 of Appendix F of the Victorian Electricity Distribution Network Service Providers Distribution Determination 2011-2015. This formula includes the calculation requirements for under and over recovery amounts from the two preceding years. JEN notes that it has only previously had jurisdictional scheme tariffs in 2010.

Attachment 1 provides the calculations demonstrating JEN's MJR_t compliance.